



**MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO**  
**DEPARTMENT OF ELECTRONIC ENGINEERING**  
**INTEGRATED ELECTRONICS # 03**



**To design and analyze the behavior of NAND and NOR Gates using  
Switches, Diodes, Transistor**

|                    |               |
|--------------------|---------------|
| <b>Roll No:</b>    | <b>Date:</b>  |
| <b>Checked by:</b> | <b>Score:</b> |

| LAB PERFORMANCE INDICATOR | SUBJECT KNOWLEDGE | DATA ANALYSIS AND INTERPRETATION | ABILITY TO CONDUCT EXPERIMENT | PRESENTATION | DESIGN AND CODING | OBSERVATION/ RESULTS | SCORE |
|---------------------------|-------------------|----------------------------------|-------------------------------|--------------|-------------------|----------------------|-------|
|                           |                   |                                  |                               |              |                   |                      |       |

**Performance:**

**Objective:** To learn the methods of designing gates using switches, diodes and transistors.

**Apparatus:**

- Diodes
- Transistors
- Resistors
- Switches (SPST)
- Multimeter
- Connecting wire

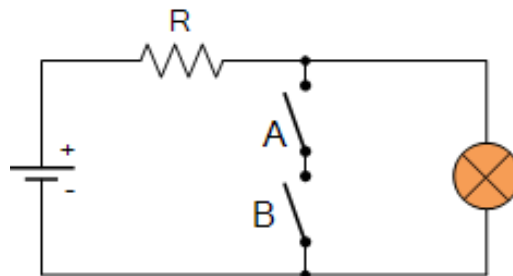
**NAND Gate:**

The NAND or Not AND function is a combination of the two separate logical functions, the AND function and the NOT function connected together in series. The **Logic NAND Function** only produces an output 1 when "ANY" of its input is 0 and in Boolean algebra terms the output will be TRUE only when any of its inputs are FALSE.



Logic Diagram

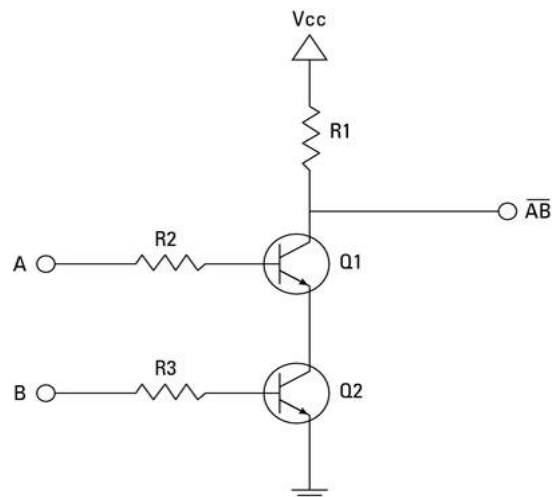
NAND gate can be formed by using switches, and transistors, circuit diagrams are given below.



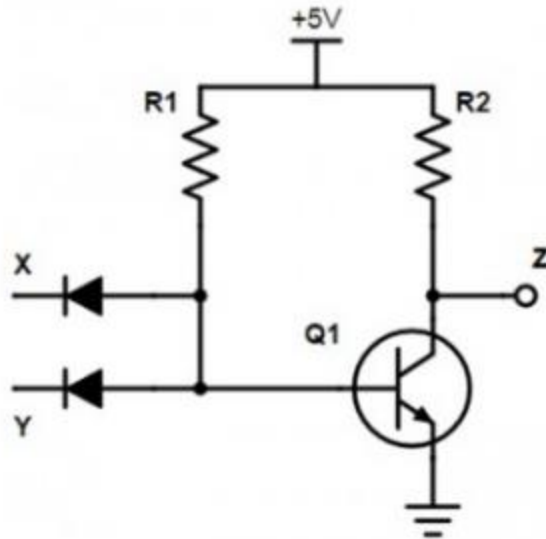
Switch A - Open = "0", Closed = "1"

Switch B - Open = "0", Closed = "1"

NAND Gate using Switches



NAND Gate using Transistors



NAND Gate using Diode Transistor Logic

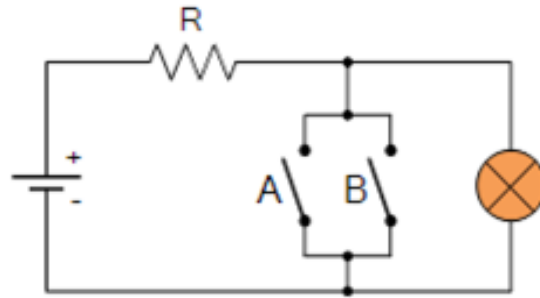
### NOR Gate:

Like the NAND Gate above, the NOR or Not OR Gate is also a combination of two separate functions, the OR function and the NOT function connected together in series and is expressed by the Boolean expression as,  $A + B$ . The **Logic NOR Function** only produces an output 1 when "ALL" of its inputs are 0 and in Boolean algebra terms the output will be TRUE only when all of its inputs are FALSE.



Logic Diagram

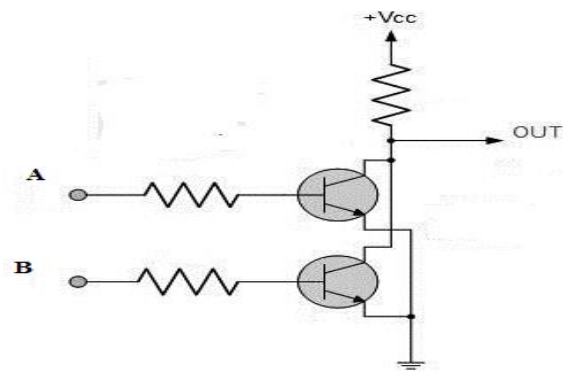
- NOR gate can be formed by using switches, and transistors, circuit diagrams are given below.



Switch A - Open = "0", Closed = "1"

Switch B - Open = "0", Closed = "1"

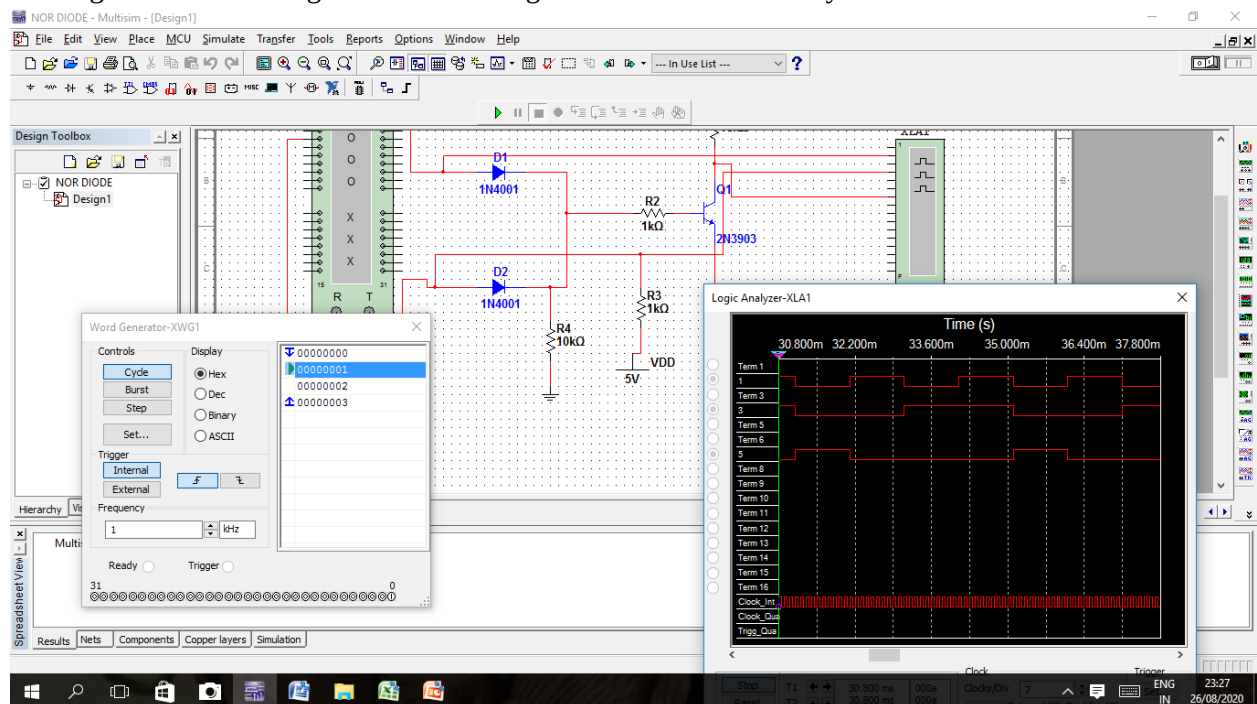
NOR Gate using Switches



NOR Gate using Transistors

**Questions:**

## 1. Design NOR Gate using diode transistor logic on Multisim and verify its truth-table?



2. Is negative supply is required to bias the diode transistor gates if yes explain what is the reasons behind that?

Yes the negative supply is required to the bias the diode transistor the reason is that because the carries takes more time for discharged and face some delays so we use negative supply then carries immediately will discharged and consume less time.

3. Write few advantages of DTL as compare to other logic families.

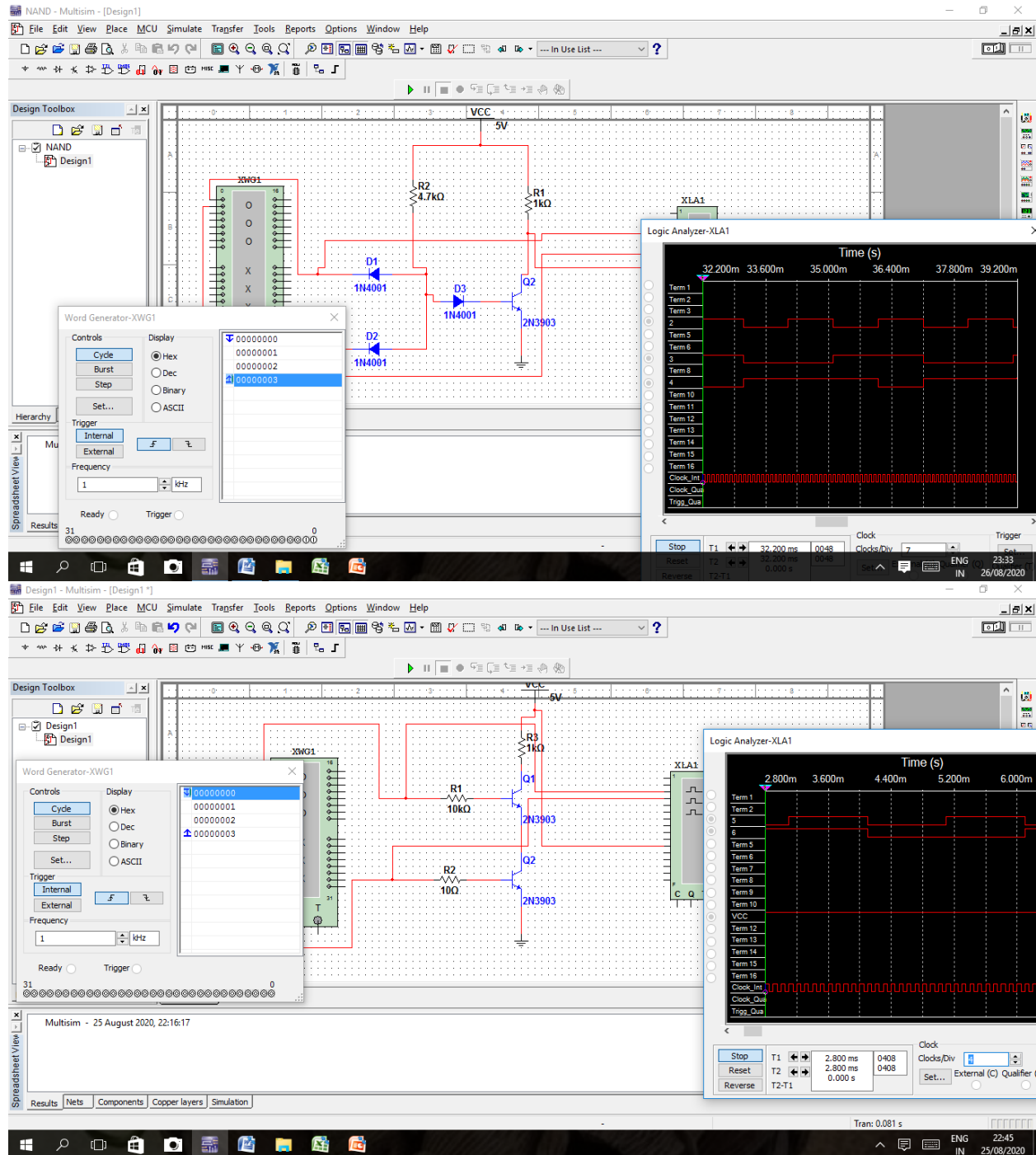
One advantage of digital circuits when compared to analog circuits is that signal represented digitally Can be transmitted without degradation due to noise has small size and it is more reliable and inexpensive than other logic families.

## Lab Task-1

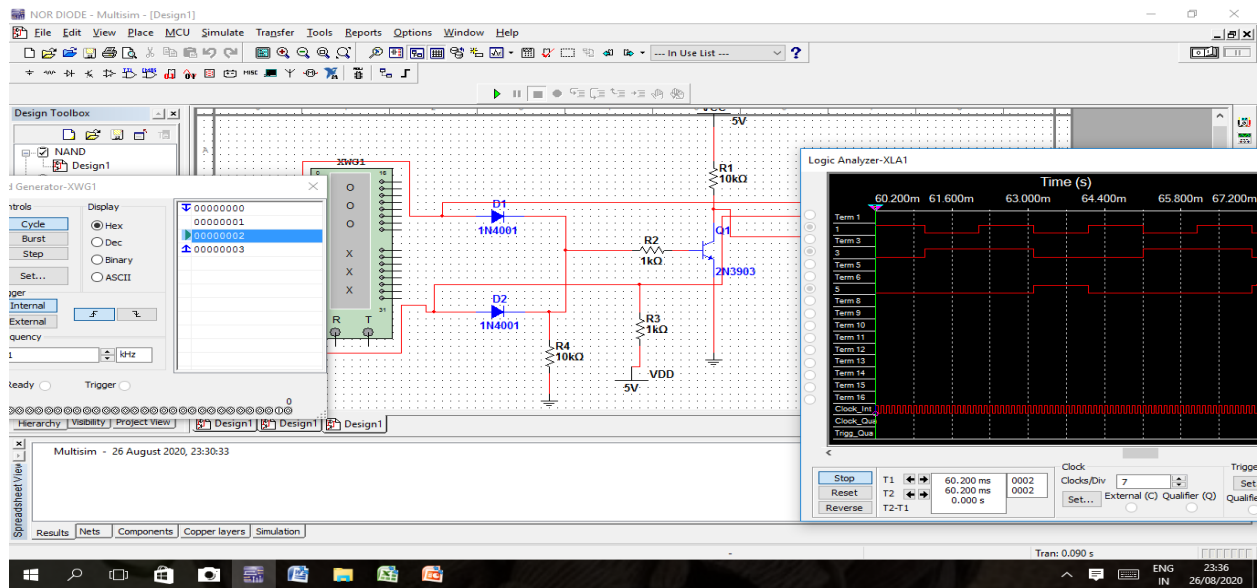
- Design the circuits in MULTISIM according to circuit diagrams given above for all Logic Gates attach the screenshots and verify truth tables.

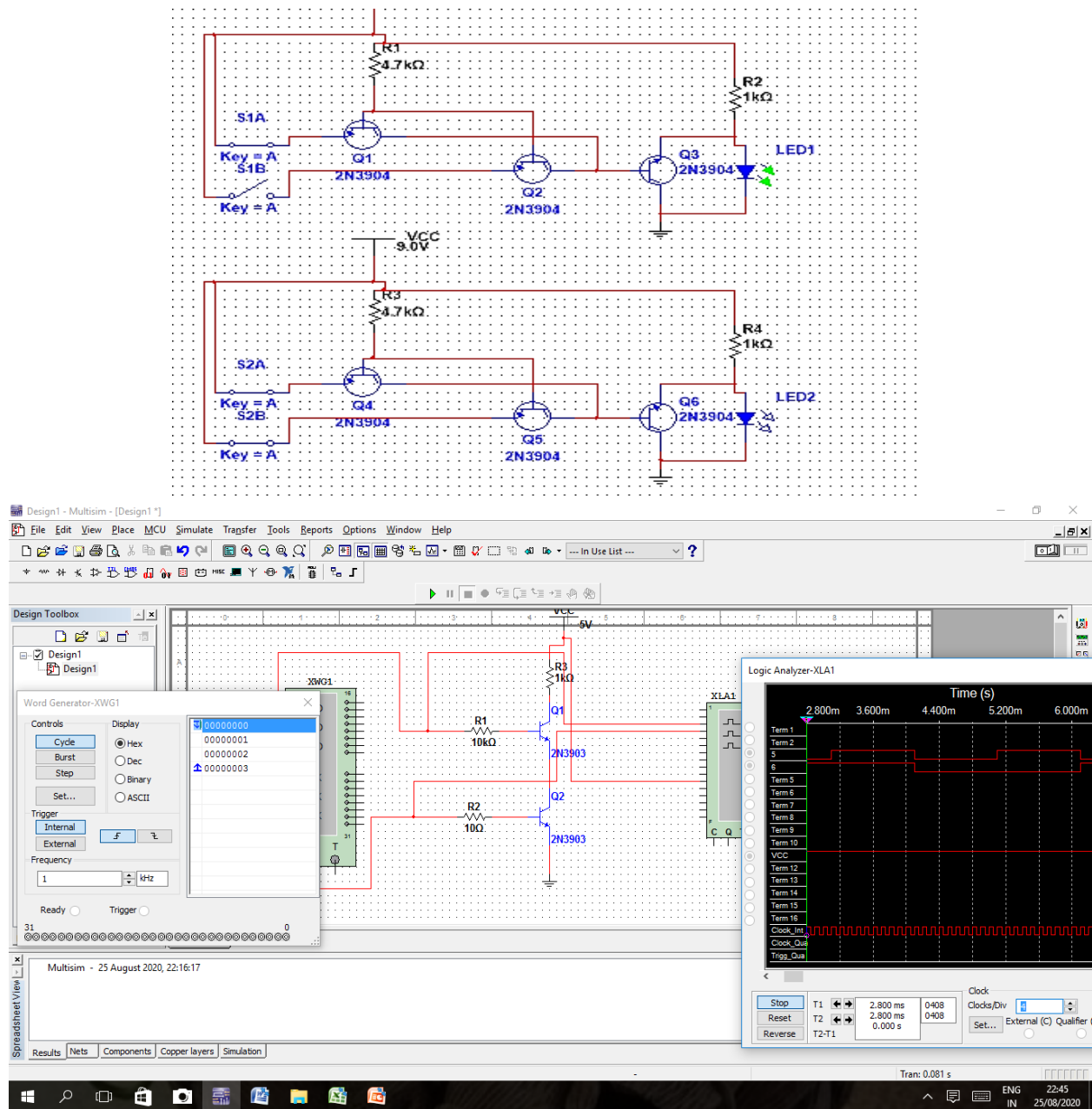
## Lab Task-2: NAND Gate

# NAND GATE

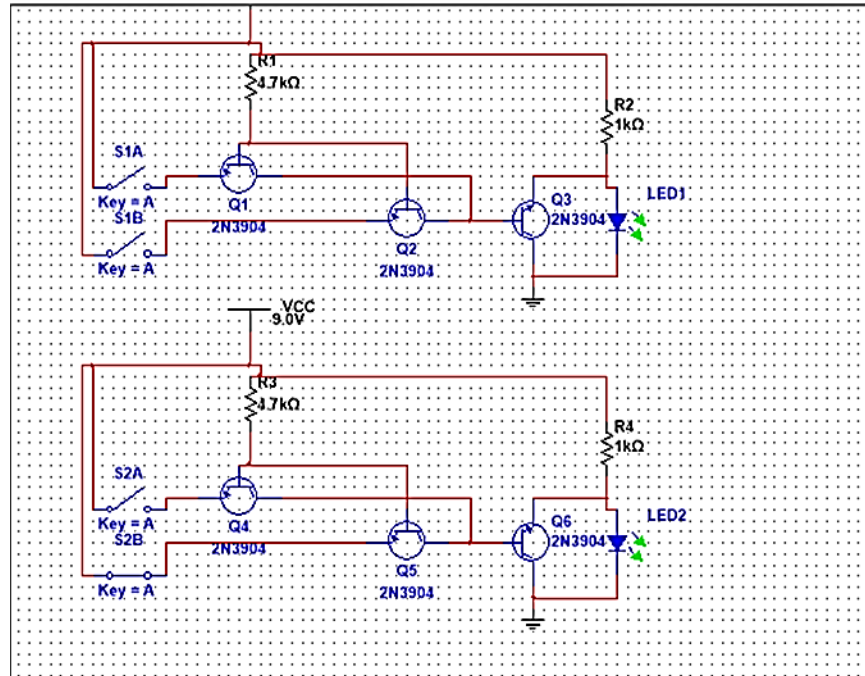


# NOR GATE

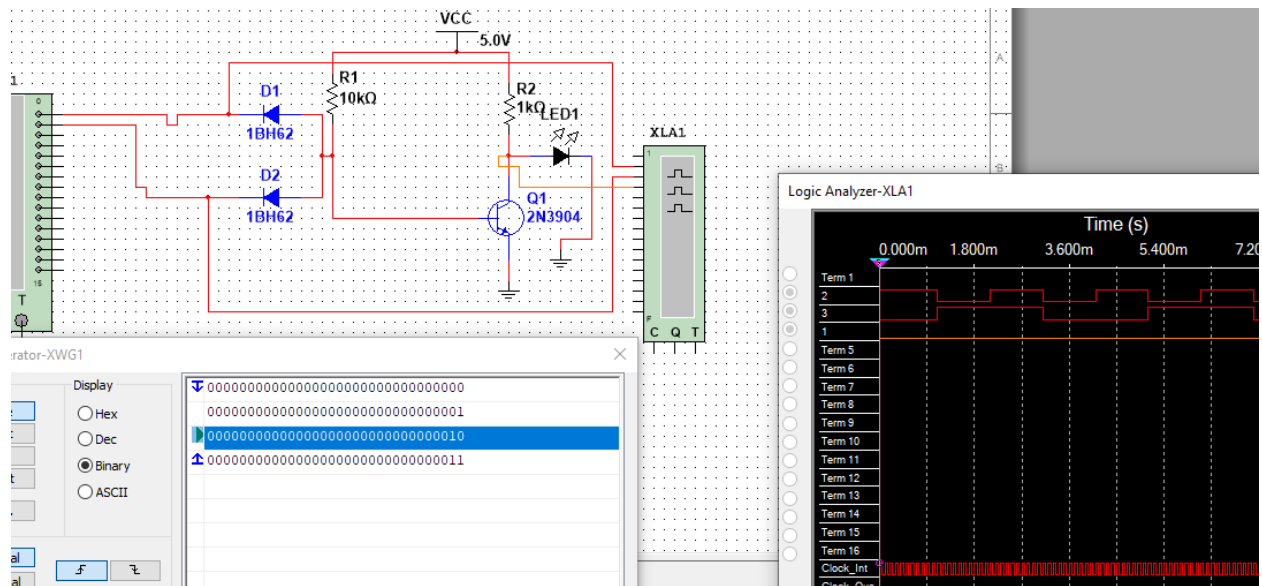








Simulation OR and AND Gate



NOR LOGIC USING DIODES